Discrete Dual Spacetime: Bounded Torsion Scalar and Finite Discrete Rotor Image

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Abstract

The dual spacetime theory rejects the continuum hypothesis entirely, positing that each massive particle carries a pair of intrinsically compactified Minkowski spacetimes encoded in the 16-real-dimensional biquaternion algebra isomorphic to $\mathrm{Cl}(3,1)$. In this companion paper, we formalize the discretization of the rapidity parameters arising from the dual map $X \mapsto Xi$. Cyclic (elliptic) rotor exponentials are 2π-periodic; hyperbolic (boost) exponentials are nowhere 2π-periodic, so placing boosts on a 2π-torus is a modelling choice rather than a consequence of the exponential map. Admissibility—non-negative embedded rapidities with the sum bound $\alpha_a + \beta_a \le \pi/2$ per axis—is an additional hypothesis, not a consequence of discretizing the torus. That sum bound is not the rotor condition $R_{\mathrm{dual}} = R^{\dagger}_{\mathrm{usual}}$, which the parent theory calls anti-synchronisation. On that cone the normalized torsion scalar satisfies $|J_{\mathrm{norm}}| \le 1$, equivalently $|J| \le 3\pi^2/8$. The unsigned mass meets the same numerical ceiling on a larger locus: every axis at a corner, of either type. The admissible locus itself is a product of three triangular cones of side $\lfloor N/4 \rfloor$, of cardinality $T(\lfloor N/4 \rfloor)^3$ with $T(K) = (K+1)(K+2)/2$. The occupied values of J_{norm} form a finite subset of the rational lattice of spacing $16/(3N^2)$: the first nonzero slot $\Delta = \pm 1$ is attained for every $N \ge 4$, so the ultraviolet floor $|J_{\mathrm{norm}}| \ge 16/(3N^2)$ is sharp, while for every $K = \lfloor N/4 \rfloor \ge 2$ a ceiling band of $2K-2$ positive integers lies strictly between the attained shoulder $3K^2 - 2K + 1$ and the ceiling $3K^2$ (at $N = 8$ the band is exactly $|\Delta| \in \{10, 11\}$). What is finite is the image of $(\mathbb{Z}/N\mathbb{Z})^6$ under the torsion-rotor map, not the unit group of an unspecified integer biquaternion order. Same-axis generators commute, so a one-axis exponential factorises. Distinct-axis generators need not: the commutator of a boost on one axis with a rotation on another does not vanish, so a generic six-parameter exponential is not justified as a product of one-axis factors. That non-commutativity does not by itself decide whether the image set equals such a product. Along multiples of 4 the occupied spectrum becomes dense in $[-1, 1]$. The admissible ceiling survives that limit. Local teleparallel equivalence belongs to the continuous theory; the ceiling is not a bound on laboratory proper acceleration, and it does not restore configurations outside the cone.

1 Introduction

The dual spacetime theory, as presented in Ref. [1], achieves an exact algebraic reformulation of General Relativity as the teleparallel equivalent (TEGR) using biquaternions, while interpreting gravity and inertia as torsional mismatch between particle-intrinsic usual and dual spacetimes. A central philosophical tenet is the complete rejection of the spacetime continuum hypothesis: no shared differentiable manifold exists; spacetime degrees of freedom are demoted to internal attributes of particles.

This rejection naturally invites a discrete structure at the fundamental scale. The biquaternion algebra $\mathbb{H} \oplus \mathbb{H} \cong \mathrm{Cl}(3,1)$ already encodes compactness through the pseudoscalar i ($i^2 = -1$) and the dual map $X \mapsto Xi$, which reverses the temporal arrow while preserving the Minkowski norm. Only the cyclic sector of the rotor exponential is intrinsically periodic; the hyperbolic sector is not.

In this paper, we develop this discretization. By placing the six rapidity coordinates on the finite torus $(\mathbb{Z}/N\mathbb{Z})^6$, where N is a large integer related to the particle's compactification scale, we obtain:

- A bounded normalized torsion scalar $|J_{\mathrm{norm}}| \le 1$ on admissible configurations (equivalently $|J| \le 3\pi^2/8$), due to the sign asymmetry of the Killing form together with the extra admissible-cone hypotheses (non-negative rapidities and the sum bound $\alpha_a + \beta_a \le \pi/2$). Discretization of the torus alone does not imply the bound. The unsigned mass shares the numerical ceiling and attains it at every corner assignment, not only at the two pure walls.

- A sharp discrete refinement $|J_{\text{norm}}| \le (4\lfloor N/4 \rfloor/N)^2$, with equality ± 1 attained if and only if $4 \mid N$. When $N < 4$ the admissible cone is a singleton and $J_{\text{norm}} = 0$.
- An explicit count of admissible points, $T(\lfloor N/4 \rfloor)^3$, and a rational spectrum $J_{\text{norm}} = 16\Delta/(3N^2)$ for an integer mismatch Δ. For $N \ge 4$ the value $\Delta = \pm 1$ is attained, so the smallest nonzero $|J_{\text{norm}}|$ is exactly $16/(3N^2)$.
- A ceiling band of spectral holes. On one axis the second maximum in the triangle is $(K-1)^2$, separated from the wall value K^2 because a positive pair on the hypotenuse $n + m = K$ satisfies $|n - m| \le K - 2$. The three-axis shoulder $S(K) = 3K^2 - 2K + 1$ is attained, and every integer strictly between $S(K)$ and $3K^2$ is empty. For $K \ge 2$ that positive band contains $2K - 2$ slots; at $N = 8$ it is exactly $|\Delta| \in \{10, 11\}$. Interior holes, such as $|\Delta| = 20$ at $K = 3$, lie below the shoulder. The normalized width of the band tends to zero along multiples of 4, so density in $[-1, 1]$ survives.
- A finite *discrete rotor image*: the image of $(\mathbb{Z}/N\mathbb{Z})^6$ under $t \mapsto \exp(\Omega_{\text{torsion}}(t))$. This is not the unit group of an integer biquaternion order, which may be infinite (Pell-type). Same-axis generators commute, so a one-axis factorisation is justified; a boost and a rotation on distinct axes do not commute, so the same factorisation is not justified off axis. Non-commutativity does not by itself identify the image with, or separate it from, a Cartesian product of one-axis images.
- A finite admissible search space. Quantizing hyperbolic rapidities onto a 2π-torus is a modelling choice. Real-scale admissibility forces the total modular winding to vanish, so a winding obstruction cannot fire on a seed whose scaling stays in the cone. Balance is not the reason: the same vanishing holds for every scale-admissible seed.

Along multiples of 4, the occupied spectrum becomes dense in $[-1, 1]$ as $N \to \infty$. The admissible ceiling survives the limit. Local teleparallel equivalence is a statement of the continuous theory. Configurations outside the cone are not restored.

2 Review of Continuous Dual Spacetime

Each particle carries usual spacetime vectors $X = ctj + xkI + ykJ + zkK$ and dual vectors $X' = ct'k + x'jI + y'jJ + z'jK$, related by $X' = Xi$. The complete rotor is

$$R_{\text{total}} = \exp\left(\sum_{a=1}^{3} \frac{\omega_a}{2} i\Gamma_a + \frac{\phi_a}{2}\Gamma_a\right), \quad \Gamma_1 = I,\ \Gamma_2 = J,\ \Gamma_3 = K.$$

The torsional mismatch rotor $\Omega = R_{\text{usual}}^{\dagger} R_{\text{dual}}$ yields the bivector $\Omega_{\text{biv}} = \log \Omega$, and the scalar

$$J = \frac{1}{2}\sum_{a=1}^{3}(\alpha_a^2 - \beta_a^2),$$

with $J_{\text{norm}} = \frac{8}{3\pi^2} J$. Under the half-angle expansion $\Omega_{\text{biv}} = \sum_a \left(\frac{\alpha_a}{2} i\Gamma_a + \frac{\beta_a}{2}\Gamma_a\right)$ and $B(i\Gamma_a, i\Gamma_a) = +8$, $B(\Gamma_a, \Gamma_a) = -8$, one has $B(\Omega_{\text{biv}}, \Omega_{\text{biv}}) = 2\sum_a(\alpha_a^2 - \beta_a^2)$, so J is four times $\frac{1}{16} B(\Omega_{\text{biv}}, \Omega_{\text{biv}})$. The six generators span a Lorentz candidate $\mathfrak{so}(3,1)$, not $\mathfrak{so}(3,1) \oplus \mathfrak{so}(3,1)$.

The same quadratic data determine an unsigned mass

$$M = \frac{1}{2}\sum_{a=1}^{3}(\alpha_a^2 + \beta_a^2), \qquad M_{\text{norm}} = \frac{8}{3\pi^2} M.$$

Three vanishing conditions must not be conflated. Vanishing of M forces vanishing of J, and that common vanishing is the vacuum. The converse is false: the balanced ray $\alpha_a = \beta_a$ has $J_{\text{norm}} = 0$ with $M_{\text{norm}} > 0$ whenever the common rapidity is nonzero. Vanishing of J does not force the relative rotor to be the identity either. On one axis a balanced seed $\alpha = \beta \neq 0$ has $\Omega \neq 1$. Free fall is $\Omega = 1$, which by unitarity holds if and only if the two rotors coincide. The rotor condition $R_{\text{dual}} = R_{\text{usual}}^{\dagger}$ is a further condition: on one axis it forces both bivector coefficients to vanish, so a nontrivial pure boost does not satisfy it. Swapping the two sectors, $\alpha \leftrightarrow \beta$, sends $J \mapsto -J$ and preserves M.

3 Discretization of Rapidity Parameters

The dual map $X \mapsto Xi$ implies an intrinsic identification of the time direction with a circle: successive applications of the dual map twice return a sign flip ($Xii = -X$), suggesting compactification. This compactness of the dual identification must not be confused with periodicity of every rotor exponential.

Theorem 3.1 (Sectorial periodicity)**.** *Let B_a^- be a cyclic generator ($B_a^{-2} = -1$) and B_a^+ a hyperbolic generator ($B_a^{+2} = +1$). Then, for every real θ and every $k \in \mathbb{Z}$,*

$$\exp\big((\theta + 2\pi k)B_a^-\big) = \exp(\theta B_a^-),$$

while

$$\exp\big((\theta + 2\pi)B_a^+\big) \neq \exp(\theta B_a^+).$$

In particular $\exp(2\pi B_a^-) = 1$ *and* $\exp(2\pi B_a^+) \neq 1$.

Thus only the elliptic (dual-rotation) sector is intrinsically 2π-periodic. Placing hyperbolic rapidities on a 2π-torus remains a modelling choice, justified by the dual-map compactification rather than by the exponential of a boost.

We nevertheless discretize both sectors uniformly,

$$\omega_a = \frac{2\pi n_a}{N}, \quad \phi_a = \frac{2\pi m_a}{N}, \quad n_a, m_a \in \mathbb{Z}/N\mathbb{Z},$$

where $N \gg 1$ is the effective number of states in the compact direction. Physically, $N \propto mc^2/E_0$ with E_0 a fundamental energy (Planckian or particle-specific). Canonical representatives satisfy $0 \leq n_a, m_a < N$, hence the embedded rapidities lie in $[0, 2\pi)$.

Proposition 3.2. *The discrete phase space $(\mathbb{Z}/N\mathbb{Z})^6$ has cardinality N^6.*

The mismatch rotor Ω therefore takes values in a finite set, namely the image of this torus under the torsion-rotor exponential. Every discrete rotor is unitary: $R\,\widetilde{R} = 1$.

4 Bounded Torsion Scalar on Admissible Configurations

Definition 4.1 (Admissible configuration)**.** A discrete configuration is *admissible* when the embedded rapidities satisfy $\alpha_a, \beta_a \geq 0$ and $\alpha_a + \beta_a \leq \pi/2$ for each axis $a = 1, 2, 3$ (see Ref. [2] for the surrounding discussion).

Admissibility is an extra hypothesis: it is not implied by placing the six rapidities on $(\mathbb{Z}/N\mathbb{Z})^6$. Non-negativity of the embedding is automatic for canonical torus representatives, so the content of the hypothesis is the sum bound. The parent theory reserves anti-synchronisation for the rotor condition $R_{\text{dual}} = R^{\dagger}_{\text{usual}}$. On one axis that condition forces both bivector coefficients to vanish, so it is not this inequality. Equivalently, on each axis,

$$4(n_a + m_a) \leq N. \tag{1}$$

The principal-branch condition $|\alpha_a + \beta_a| \leq \pi/2$ alone does not bound J.

The Killing form asymmetry implies that maximal attraction occurs when usual boosts dominate ($\alpha_a \gg \beta_a$), and maximal repulsion when dual rotations dominate. On the continuous admissible cone the extremal values of the mismatch are pure hyperbolic dominance ($\alpha_a = \pi/2$, $\beta_a = 0$, all a), giving $J = \frac{3\pi^2}{8}$ and $J_{\text{norm}} = 1$, or pure elliptic dominance ($\alpha_a = 0$, $\beta_a = \pi/2$), giving $J = -\frac{3\pi^2}{8}$ and $J_{\text{norm}} = -1$. Equality $|J_{\text{norm}}| = 1$ holds if and only if every axis is of the same extremal type. The unsigned mass obeys the same numerical ceiling, $0 \leq M_{\text{norm}} \leq 1$, on a larger locus: $M_{\text{norm}} = 1$ if and only if every axis is a corner, a pure boost or a pure rotation, independently of the other axes. If n axes are pure boosts, the height at that mass ceiling is $(2n - 3)/3$, hence one of 1, $1/3$, $-1/3$, -1. A balanced configuration does not meet the mass ceiling: on the cone, $J_{\text{norm}} = 0$ forces $M_{\text{norm}} \leq 5/6$.

Theorem 4.2 (Continuous admissible bound)**.** *For every admissible continuous configuration,*

$$|J| \leq \frac{3\pi^2}{8}, \qquad |J_{\text{norm}}| \leq 1, \qquad 0 \leq M_{\text{norm}} \leq 1.$$

The image of J_{norm} on that cone is exactly $[-1, 1]$.

Configurations that violate admissibility (e.g. negative embedded rapidities while $|\delta_a| \leq \pi/2$) need not satisfy this bound.

4.1 Integer mismatch and the discrete spectrum

Write

$$\Delta(t) = \sum_{a=1}^{3} \big(n_a^2 - m_a^2\big), \qquad \Sigma(t) = \sum_{a=1}^{3} \big(n_a^2 + m_a^2\big)$$

for a torus point $t = (n_a, m_a)$. The torsional scalars are then exactly the rational multiples

$$\begin{aligned} J &= \tfrac{1}{2}\big(2\pi/N\big)^2 \Delta(t), \quad J_{\mathrm{norm}} = \frac{16}{3N^2}\Delta(t), \\ M &= \tfrac{1}{2}\big(2\pi/N\big)^2 \Sigma(t), \quad M_{\mathrm{norm}} = \frac{16}{3N^2}\Sigma(t). \end{aligned} \tag{2}$$

In particular $J_{\mathrm{norm}} = 0$ if and only if $\Delta(t) = 0$. Balanced axes $n_a = m_a$ give a sufficient condition for vanishing J, but not a necessary one: the cross-axis cancellation $(n, m) = (1, 0), (0, 1), (0, 0)$ is admissible for every $N \geq 4$, has $\Delta = 0$ and $\Sigma = 2$, and is not balanced. Unsigned mass and torsion occupy the even sublattice

$$|\Delta| \leq \Sigma \leq 3\big\lfloor N/4 \big\rfloor^2, \qquad \Sigma \equiv \Delta \pmod 2.$$

Vacuum is $\Sigma = 0$, hence only the origin. Vanishing J with positive mass is $\Delta = 0$ and $\Sigma > 0$.

The one-axis mismatches are the differences of squares $n^2 - m^2$ in the triangle $n + m \leq K = \lfloor N/4 \rfloor$. The three-axis image of Δ is the corresponding three-fold sumset, a subset of $\{-3K^2, \ldots, 3K^2\}$. For $N \geq 4$ one has $K \geq 1$, so $\Delta = \pm 1$ is attained (one axis at $(1, 0)$ or $(0, 1)$, the others at the origin). Consequently the universal lower bound on nonzero height is sharp:

$$\min\big\{|J_{\mathrm{norm}}| : \text{ admissible, } \Delta \neq 0\big\} = \frac{16}{3N^2} \qquad (N \geq 4). \tag{3}$$

The bound interval is not fully occupied. The integers immediately beneath the ceiling are empty for a uniform reason, recorded once that ceiling is identified.

Theorem 4.3 (Sharp discrete bound)**.** *On every admissible discrete configuration,*

$$\big|\Delta(t)\big| \leq 3\big\lfloor N/4 \big\rfloor^2, \qquad \Sigma(t) \leq 3\big\lfloor N/4 \big\rfloor^2,$$

and therefore

$$|J_{\mathrm{norm}}| \leq \Big(\frac{4\lfloor N/4 \rfloor}{N}\Big)^2, \qquad M_{\mathrm{norm}} \leq \Big(\frac{4\lfloor N/4 \rfloor}{N}\Big)^2 \leq 1.$$

If $4 \mid N$, both signs $J_{\mathrm{norm}} = \pm 1$ are attained, at the pure-hyperbolic lattice point $n_a = N/4$, $m_a = 0$ and the pure-elliptic point $n_a = 0$, $m_a = N/4$ respectively. If $4 \nmid N$, every admissible point satisfies $|J_{\mathrm{norm}}| < 1$. Equality $|J_{\mathrm{norm}}| = (4\lfloor N/4 \rfloor/N)^2$ holds if and only if all three axes sit on the same *wall: all $(K, 0)$ or all $(0, K)$.*

The mass ceiling has a different locus. On one axis, $n^2 + m^2 = (n+m)^2 - 2nm$, so inside the triangle the value K^2 is attained only when the cross term vanishes and the pair sits on the hypotenuse, that is only at a wall corner.

Theorem 4.4 (Mass ceiling locus)**.** *Let $K = \lfloor N/4 \rfloor$. On the triangle $n + m \leq K$, the axis mass $n^2 + m^2$ equals K^2 if and only if (n, m) is $(K, 0)$ or $(0, K)$. Consequently $\Sigma(t) = 3K^2$ if and only if every axis is a wall corner, of either type. For $K \geq 1$ there are eight such corners. A mixed corner, two axes at $(K, 0)$ and one at $(0, K)$, attains $\Sigma = 3K^2$ and satisfies $|\Delta| = K^2 < 3K^2$. Mixed walls therefore meet the mass ceiling and lie strictly inside the torsion bound.*

The same-wall criterion leaves open how close a configuration may approach the ceiling without attaining it. The one-axis mismatch factors as $n^2 - m^2 = (n - m)(n + m)$. On the triangle $n + m \leq K$ the product is at most $(n + m)^2 \leq K^2$, with equality only at a pure wall. If the pair is not a wall, either $n+m \leq K-1$, in which case the product is at most $(K-1)^2$, or the pair lies on the hypotenuse $n+m = K$ with both coordinates positive. There $|n - m| \leq K - 2$, so the product is at most $K(K - 2) = K^2 - 2K$, which sits strictly below $(K - 1)^2 = K^2 - 2K + 1$. The open interval between the second maximum $(K-1)^2$, attained at the shortened wall $(K - 1, 0)$, and the wall value K^2 is therefore empty on a single axis.

A three-axis sum can exceed the shoulder

$$S(K) = 2K^2 + (K - 1)^2$$

only by placing every axis on the same pure wall. For $K \geq 1$ this shoulder is the integer $S(K) = 3K^2 - 2K + 1$. The pure-wall configuration jumps from $S(K)$ to the ceiling $3K^2$ and skips every intermediate integer. The shoulder itself is attained, by two axes at $(K, 0)$ and one axis at $(K-1, 0)$, and likewise by the sign-reversed elliptic walls.

Theorem 4.5 (Ceiling band)**.** *Let $K = \lfloor N/4 \rfloor$. Every integer Δ with $S(K) < |\Delta| < 3K^2$ is unoccupied by admissible configurations. For $K \geq 2$ there are exactly $2K - 2$ such positive integers. For $K = 1$ the open interval $(S(1), 3)$ contains no integer, in agreement with the full occupation of $\{-3, \ldots, 3\}$.*

Thus the holes $|\Delta| \in \{10, 11\}$ at $N = 8$ ($K = 2$, $S(2) = 9$) are the entire ceiling band. At $N = 12$ ($K = 3$, $S(3) = 22$) the same band accounts for $|\Delta| \in \{23, 24, 25, 26\}$; the further vacancy $|\Delta| = 20$ lies strictly below the shoulder and is an interior hole, not forced by the isolation of the wall. In the normalized scalar the distance from the shoulder to the ceiling is $(2K - 1) \cdot 16/(3N^2)$. Along multiples of 4, where $K = N/4$, this distance equals $(8N - 16)/(3N^2)$ and tends to zero, so the band does not obstruct density of the occupied spectrum in $[-1, 1]$. At each finite N the discrete spectrum nevertheless retains holes: the ceiling band for every $K \geq 2$, together with interior holes such as $|\Delta| = 20$ at $K = 3$.

Corollary 4.6. *If $N < 4$, the only admissible point is the origin, and $J_{\text{norm}} = 0$ identically on the admissible cone.*

For $N \geq 8$ the balanced lattice point $n_a = m_a = 1$ (all axes) is admissible, has $\Delta = 0$, and has strictly positive mass. The balanced cube itself has cardinality $(\lfloor K/2 \rfloor + 1)^3$; the zero-height locus is strictly larger once $K \geq 1$ (seven points at $N = 4$, thirty-two at $N = 8$, sixty-eight at $N = 12$). Vanishing J on the discrete cone does not force the vacuum.

4.2 Cardinality of the admissible cone

Inequality (1) factorizes across axes. On each axis the pairs (n, m) with $n + m \leq \lfloor N/4 \rfloor$ form a triangular region of cardinality $T(K) = (K + 1)(K + 2)/2$ with $K = \lfloor N/4 \rfloor$.

Theorem 4.7 (Admissible cardinality)**.** *The number of admissible configurations is*

$$T\big(\lfloor N/4 \rfloor\big)^3 = \left(\frac{\big(\lfloor N/4 \rfloor + 1\big)\big(\lfloor N/4 \rfloor + 2\big)}{2} \right)^3 .$$

In particular the count depends on N only through $\lfloor N/4 \rfloor$. For $N < 4$ it equals 1; for $N = 4$ it equals 27; for $N = 8$ it equals 216.

The admissible values of J_{norm} therefore form a finite subset of the scaled integer lattice $\frac{16}{3N^2}\mathbb{Z}$, lying in the interval determined by Theorem 4.3, and occupying only those slots whose integer mismatch lies in the three-fold sumset. As $N \to \infty$ through multiples of 4, this discrete image becomes dense in $[-1, 1]$: every value in $[-1, 1]$ is approximated arbitrarily closely by J_{norm} of an admissible torus point. The ceiling band of Theorem 4.5 shrinks in the normalized scalar, but at each finite N it, and any interior holes, remain empty.

In the continuum limit the admissible envelope $\alpha_a, \beta_a \in [0, \pi/2]$ with $\alpha_a + \beta_a \leq \pi/2$ persists; the bound $|J_{\text{norm}}| \leq 1$ therefore remains valid in the limiting theory on that domain, while TEGR equivalence is recovered locally.

5 Integer Biquaternion Ring and Finite Rotor Image

Define the integer biquaternion ring as the $\mathbb{Z}$-span of the basis with Gaussian/Hurwitz integer coefficients in each quaternion sector (precise order to be classified). That order is not constructed here, and a Lorentzian integer order may have infinitely many units (Pell-type).

What is finite—and all that the discrete torus supplies—is the *discrete rotor image*: the image of $(\mathbb{Z}/N\mathbb{Z})^6$ under the torsion-rotor map $t \mapsto \exp(\Omega_{\text{torsion}}(t))$. Finiteness of the domain implies finiteness of the image; no enumeration, and no identification with $R^\times$ of an integer order, is required. The image of the admissible subclass is a finite subset of that rotor image.

Same-axis generators commute, so a one-axis mixed exponential factorises as a product of a hyperbolic rotor and an elliptic rotor, with closed forms cosh / sinh and cos / sin. The commutator of the boost on axis 0 with the rotation on axis 1 does not vanish. A generic six-parameter exponential is therefore not

justified as a product of three one-axis factors. Whether the image set equals the Cartesian product of the three one-axis images is not settled by that commutator.

Quantizing hyperbolic rapidities onto a 2π-torus remains a modelling choice, as Theorem 3.1 records. Real-scale admissibility forces the total modular winding to vanish: if the k-fold scaling of a lattice seed stays in the continuous cone, every coordinate lies below the wrap threshold. A winding obstruction on that torus cannot fire on such a seed. The hypothesis is scale-admissibility, not balance.

6 Diophantine Constraints on Gravitational Solutions

Zero torsional height on the discrete cone is the integer condition $\Delta(t) = 0$. The admissible search space of Theorem 4.7 is finite, of cardinality $T(\lfloor N/4 \rfloor)^3$, and it sits inside the finite discrete rotor image. That finiteness turns a search over admissible torsional configurations into a finite search. It is not a derivation of the field equations from an action, and it does not identify spherically symmetric solutions with integer points on an elliptic curve. The sign of J is the sign of Δ. Finiteness of the admissible classes at each N is a theorem; a numerical nuclear cutoff is not.

7 Physical and Observational Implications

- **Expressed ceiling**: On the admissible cone $|J_{\text{norm}}| \leq 1$, equivalently $|J| \leq 3\pi^2/8$. At finite N the effective ceiling is the stricter value $(4\lfloor N/4 \rfloor/N)^2$. The torsion scalar attains it only at the same-wall lattice points; the unsigned mass attains it at every corner assignment of Theorem 4.4. The ceiling limits the mismatch a configuration expresses. Source bivectors add with no ceiling, and the expressed point is the admissible configuration closest to that sum, so its $|J|$ stays at most $3\pi^2/8$ while the same quadratic on the internal sum need not. The ceiling is not a bound on laboratory proper acceleration: surrounding matter fixes the expressed point, not the stiffness. The smallest nonzero torsional quantum on the cone is $\varepsilon_N = 16/(3N^2)$ for every $N \geq 4$.

- **Discrete spectra**: Gravitational wave echoes and nuclear energy levels acquire discrete torsional contributions on the occupied slots of the lattice (2). Not every multiple of ε_N inside the bound is realised. Immediately below the ceiling, Theorem 4.5 leaves a band of $2K - 2$ empty slots on each side, of normalized width $(2K - 1) \cdot 16/(3N^2)$. That width vanishes along multiples of 4, while the missing slots themselves remain part of the finite-N spectrum. Interior holes below the shoulder are independent of this band.

- **Nuclear chart**: Finiteness of the discrete rotor image implies only that there are finitely many admissible rotor classes at each N, namely at most $T(\lfloor N/4 \rfloor)^3$ of them. A numerical cutoff $A \lesssim 300$ is an underived heuristic, not a theorem of the discrete algebra.

- **Experimental tests**: Search for discrete torsional echoes in LIGO/Virgo data; precision clock networks probing planetary cores for exact $J = 0$ cancellation. A $J = 0$ detection would not by itself imply vanishing mass.

8 Conclusions and Outlook

The discretization yields a bounded torsion scalar on the admissible cone, a sharp discrete torsion ceiling governed by $\lfloor N/4 \rfloor$ and attained only at same-wall configurations, a mass ceiling attained at every corner assignment, an explicit triangular count of admissible points, a universal attained floor $\varepsilon_N = 16/(3N^2)$, a three-fold sumset spectrum whose ceiling is isolated from the attained shoulder $S(K) = 3K^2 - 2K + 1$ by a band of $2K - 2$ empty positive slots whenever $K \geq 2$, and a finite discrete rotor image. Same-axis exponentials factorise; off-axis factorisation is not justified by the commutators, and the image is not thereby identified with a product of one-axis images. Cyclic rotor exponentials are 2π-periodic; hyperbolic ones are not, so torus quantization of boosts remains a modelling choice. Real-scale admissibility, not balance, is what kills modular winding. The occupied spectrum is dense in $[-1, 1]$ along multiples of 4, and the ceiling survives that limit. Future work includes constructing an actual integer order (whose unit group need not be finite). Admissibility, the rotor-image finiteness, and any nuclear-chart cutoff remain logically independent of one another.

References

[1] Hypernumbernet Collaboration, "Double Spacetime Theory: Gravity as Torsion between Usual and Dual Spacetime," (2025).

[2] Hypernumbernet Collaboration, "Discrete Biquaternionic Double Spacetime: Rigorous Proofs of Diophantine Conjectures via Torsional Mismatch and PGA Embedding," (2025).